
MODULE-3

DESIGN FOR IT

Design Thinking to Business Process modeling, Agile in Virtual collaboration environment – Scenario based Prototyping

In simple terms, Design Thinking is a framework that puts user needs at the core of any product development.

The IT Industry today, is trying to build strong relations with their customers and putting efforts to receive ongoing feedback from them, while design thinking with its user-centered approach helps businesses to achieve their goals. Product developers, be it from any industry, focus first on users and their needs and elaborate solutions to effectively meet those needs.

This approach focuses on a solution rather than a problem. Relevant advice: put yourself in your clients' shoes.

Design thinking is all about iterative processing when you test one idea and see whether it is a Solution that will address the users' problem. But first things first: what exactly is the design thinking process?

Design Thinking Is Not About Design Only

The word 'design' in the name of the concept design thinking can be a little misleading. Design thinking may refer to any area: from the manufacturing industry to software development – and is appropriate for all stages of building a product.

Let's take the example of the software development process. Design thinking, in easy terms, can be applied at each stage of the software development lifecycle, be it planning, development, or testing, including design. But why is design thinking so vital for software product design and development? Let's analyze design thinking at a stage of the software development life cycle.

Design Thinking at the software designing stage

Have you ever thought why do IT industry professionals try so hard to make each detail in product design perfect and think over the colors of Call to action buttons?

The prime example that is mentioned above is the color of a button. “But this is a very minor detail!” you might say. In practice, this is an important factor since it determines whether users will use your product or not.

Sometimes users cannot instantly find a button they are looking for and then two scenarios can come into the picture:

- 1) annoyed, they close your app;
- 2) they spend a lot of time trying to find the button they need. Both the cases are not at all good for you. And both of them means that the customer’s problem was not taken into account at the design stage.

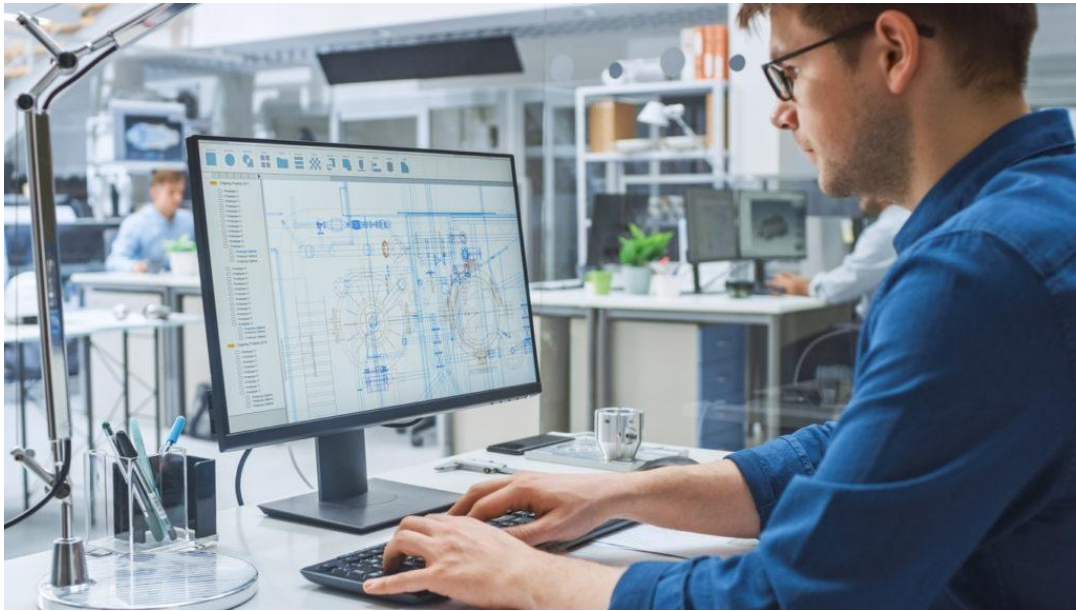
Design thinking at this stage allows the IT Industry to avoid problems like this. When you have a thorough understanding of what the customer’s problem is (the defined step) you can suggest solutions to this problem (the ideate step) and finally choose the best design and create the best possible one for all product elements. As a result, end customers do not get confused when they use your product and eventually turn into your loyal customers.

On the other hand, design thinking is a powerful tool that through specific design enables organizations to make users do what they want. All the same button colors may encourage users to take an action you need them to, for example, subscribe to the newsletter or request a trial of your product. You can look at prominent browser notifications: when, for example, a media website you visit offers to instantly notify you of important news. One button says (allow), clicking which you agree to receive push notifications. Another one is the No, thanks (block) button.

We are used to the fact that the green color usually means consent, while the color red means the dissent. As some red sign says that something is forbidden, a red button says that we disagree to receive push notifications from the website. The trick is that some media websites put the Yes on the red button. Thus, one who does not want to get bogged down in spam, intuitively clicks the red button to turn off the notifications, but, instead, starts to receive them.

Do you think it is unfair?

However, imagine that it turns out that the news we get every day/week is really useful to us but we were skeptical about it earlier. Or we start receiving weekly emails about brand-new products from our favorite online website and find the amazing offers on it! These are just a few examples that show the actual power of design thinking in software design. Speaking of buttons, color is not the one parameter that matters. Size, shape, location, order, and beyond are things to keep in mind. Buttons are just the tip of the iceberg.



Design in the IT Industry

Design Thinking at the software development stage

Now let us take a look at design thinking at another stage of the software development life cycle – development itself. Take a simple example: mobile shopping app development.

In some startups in the IT industry, where there are no UX designers and data scientists, business owners, or sometimes software developers are assumed to do their functions. To build a great product, they go deep into the problems of customers and look at each feature with users' eyes.

When it is about e-commerce shopping, it is much more convenient to use a mobile app, rather than wait for a page to load in a browser. The problem is that many e-commerce apps are a little misleading, for example, to find a product customers need, they have to wander endlessly through multiple sections and scroll down long pages.

With the help of a design thinking approach, developers pass all stages, from empathy to testing, and provide users a solution that would search for products as fast and convenient as possible. By carefully considering all the elements, developers arrange the elements logically and do everything to make the customer journey as comfortable as possible.

Benefits of design thinking in software development (IT Industry)

Below you can find the key advantages that design thinking brings to the table in the IT industry-

- Feasibility check

Design thinking can enable software development organizations to test the feasibility of the future product and its functionality at a very initial stage. With Design Thinking methods they keep end-user needs in mind, clearly specify all requirements, and translate all this into product features.

- No alarms and no surprises

Once you tested your product and gathered feedback from the customers, the team can confidently proceed to the product development. You can be quite sure that there will be little or no difference between the approved concept and the final version of the product.

- Clarity and transparency

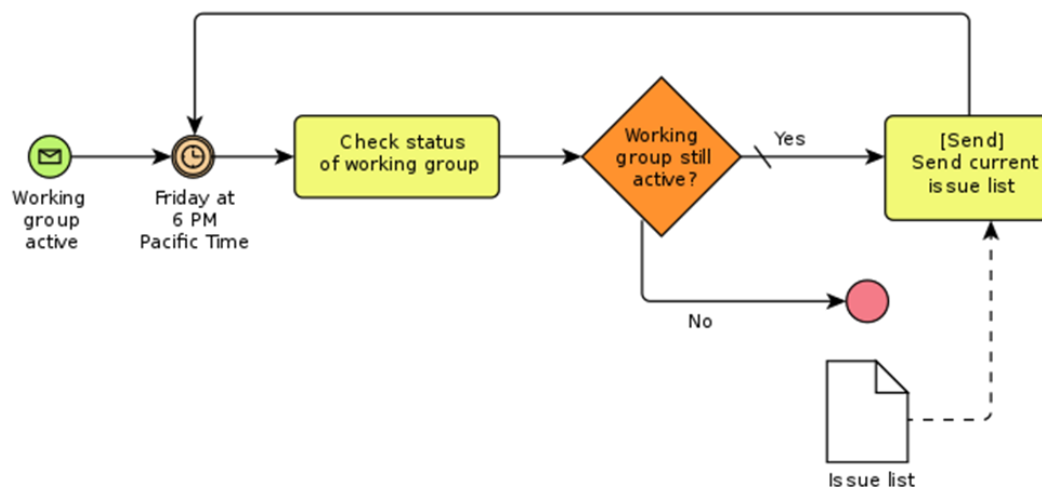
With the help of Design thinking, software developers can clearly understand the end goals and problems that users might face and build a detailed vision of the solution that they should deliver.

- Continuous improvement

The product can be modified after its release when you get the users feedback. Now you can see it very clearly: which features work and which do not; how the product can be improved and what suggestions should be incorporated. This leaves a place for continuous enhancement and the software development process becomes flexible and smooth.

DESIGN THINKING TO BUSINESS PROCESS MODELING

Business process management (BPM) is a way of breaking down business processes into their most basic elements: the tasks and activities a business carries out. BPM shows, clearly and transparently, how a product or service transforms as it moves through an organization's process sequence, often in near real-time.



Process models mediate communication between stakeholders: Business analysts, process participants and software architects. Process models provides shared understanding so that everyone can contribute knowledge

BPM – Combination of various process related steps:

- Process mapping
- Process discovery
- Process simulation
- Process analysis
- Process improvement.

BPM – Arrived from the last few decades and have replaced previous organizational efficiency packages: Time and Motion Study (TMS) and Total Quality Management (TQM). Process models mediate communication between stakeholders: Business analysts, process participants and software architects

Advantages of Business Process Management

1. Align operations with business strategy
2. Improves process communication
3. Increase control and consistency
4. Improve operational efficiencies
5. Gain competitive advantage

Design thinking in Business Process Modelling

By revealing the way things are done at an organization, and comparing that with the way things should be done, BPM highlights the dependencies and relationships between people, process, and technology—and where those elements are ripe for improvement.

Design thinking is also focused on improvement, but takes the end-user or customer experience as a starting point. Fundamentally, design thinking uses empathy to understand the way people feel about using a service or product, including where their frustrations lie, then builds on that knowledge to create improvements, with the ultimate goal of making customers' lives and experiences better and more fulfilling.

There are five key phases of design thinking:

1. There is a challenge: **How do I solve it?**
2. I have learned something: **How do I act upon it?**
3. There is an opportunity: **What do I create?**
4. There is an idea: **How do I build upon it?**
5. I tried something: **How do I evolve it?**

This is an ongoing, cyclical approach, with the practice of ‘design, test, and iterate’ at its center— that is, constantly creating and rapidly deploying prototypes or new ideas, then testing and reflecting on them, including through fast feedback from customers. Organizations therefore spend less time and resources on any one idea, and yet have the capacity to quickly scale up any idea that has merit, and positive reactions from customers.

The similarities to BPM are clear. Process improvement requires the same approach of identifying a challenge (like an inefficient process), generating the information needed to understand why the process is inefficient, implementing a possible solution, then measuring the effectiveness of that solution over time. Rinse and repeat! Merging the two disciplines essentially means using the insights gained through an effective BPM framework to feed into the design thinking process, and enhance customer-facing business processes.

After all, without a clear view of the current state operational structure and business process landscape to analyze, the risk of producing a ‘solution’ which is in fact not fit for purpose, or causes issues in other parts of the system, is amplified.

Benefits of Design Thinking and BPM

Working in tandem, design thinking and BPM unlock a range of powerful (and potentially lucrative) benefits, including:

1. Competitive advantage, gained through a willingness to innovate, then ‘double-down’ on ideas that work, and discard ones that don’t—before they become a liability
2. Moving beyond traditional process maps and case models to more customer-centric and human-scale products and services
3. Pooling collective ease and enhancing collaboration, through building multidisciplinary teams to focus on a single problem
4. Harnessing natural empathy, leading to a better understanding of the needs and challenges of customers
5. Generating more revenue by ensuring a deeper level of customer satisfaction

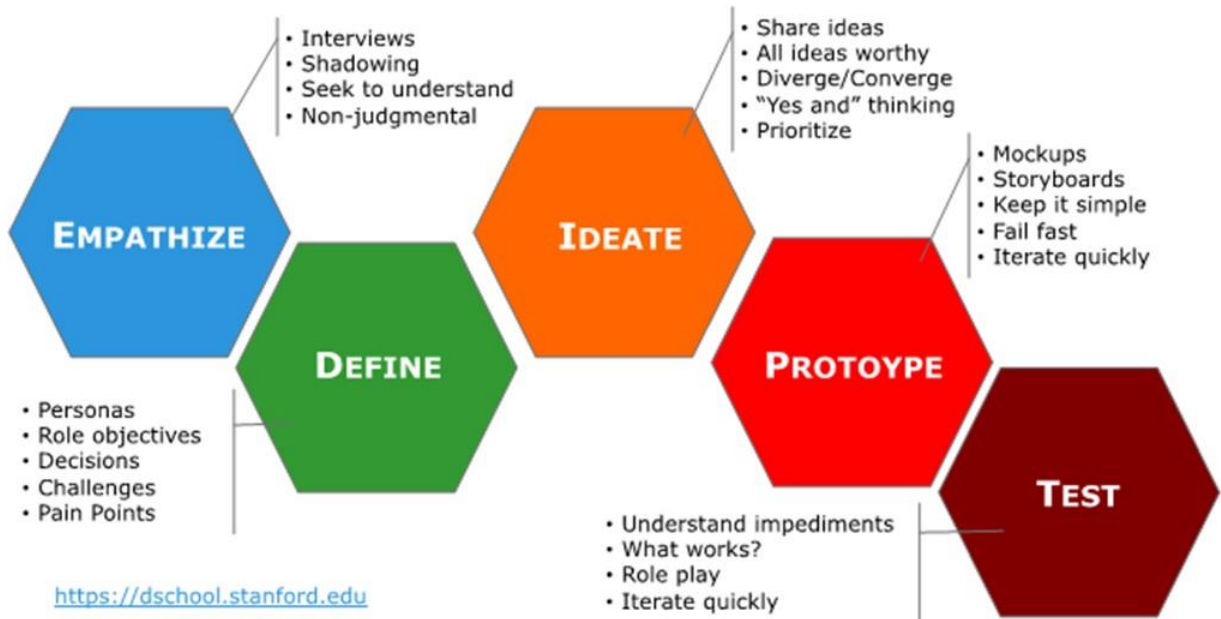
Understanding IT industry with an example

IT companies - They build software to make other businesses run smoothly!. Now if I know some technology and ask a programmer to solve problems by building specific pieces of software.

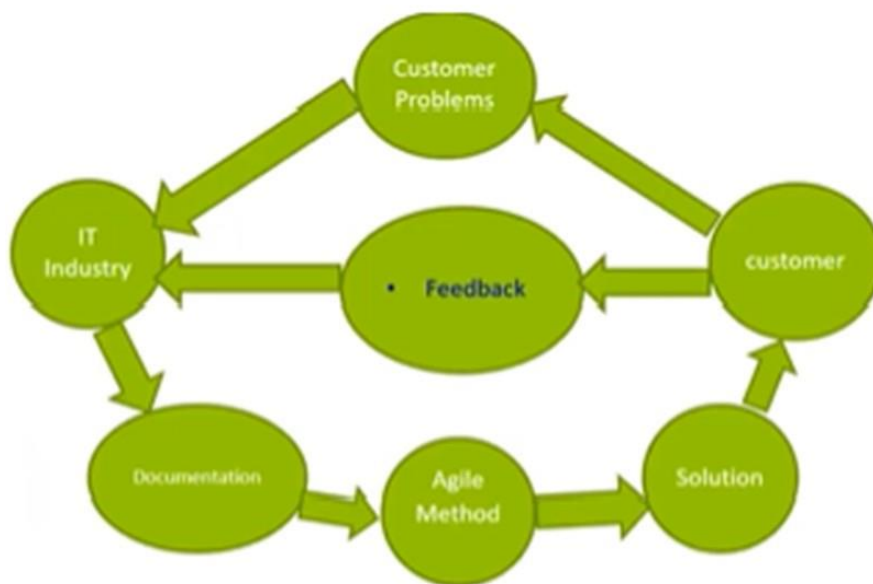
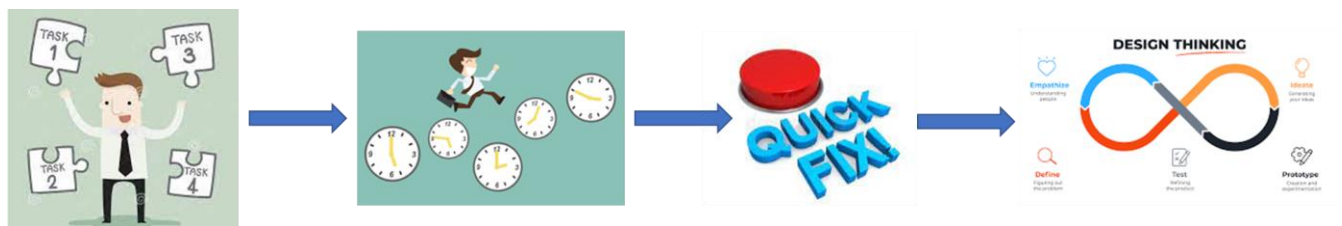
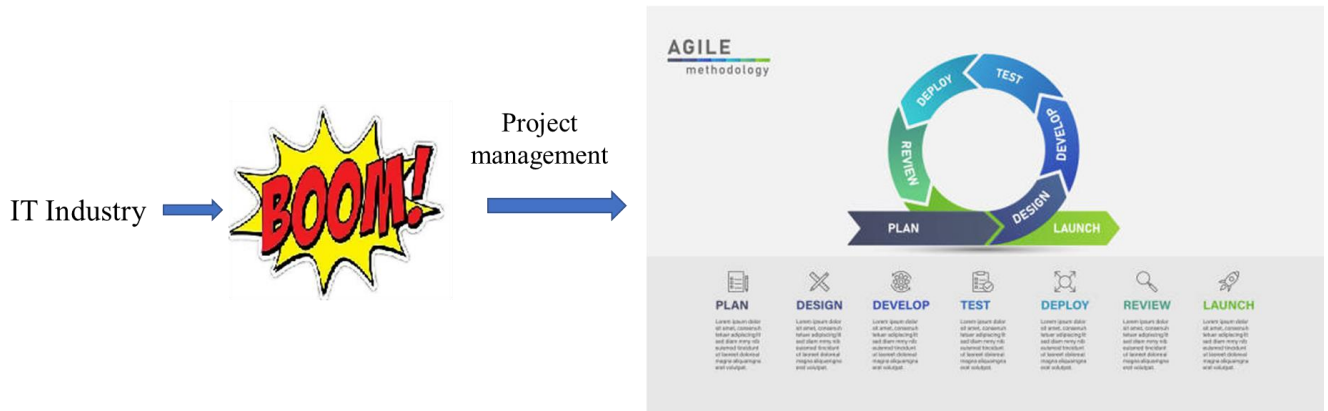


IT companies build or maintain software so that businesses run more smoothly. Anyone who builds or maintains software belongs to the IT sector.

Design Thinking in General



Design Thinking in IT industry



Agile Model of Project Management

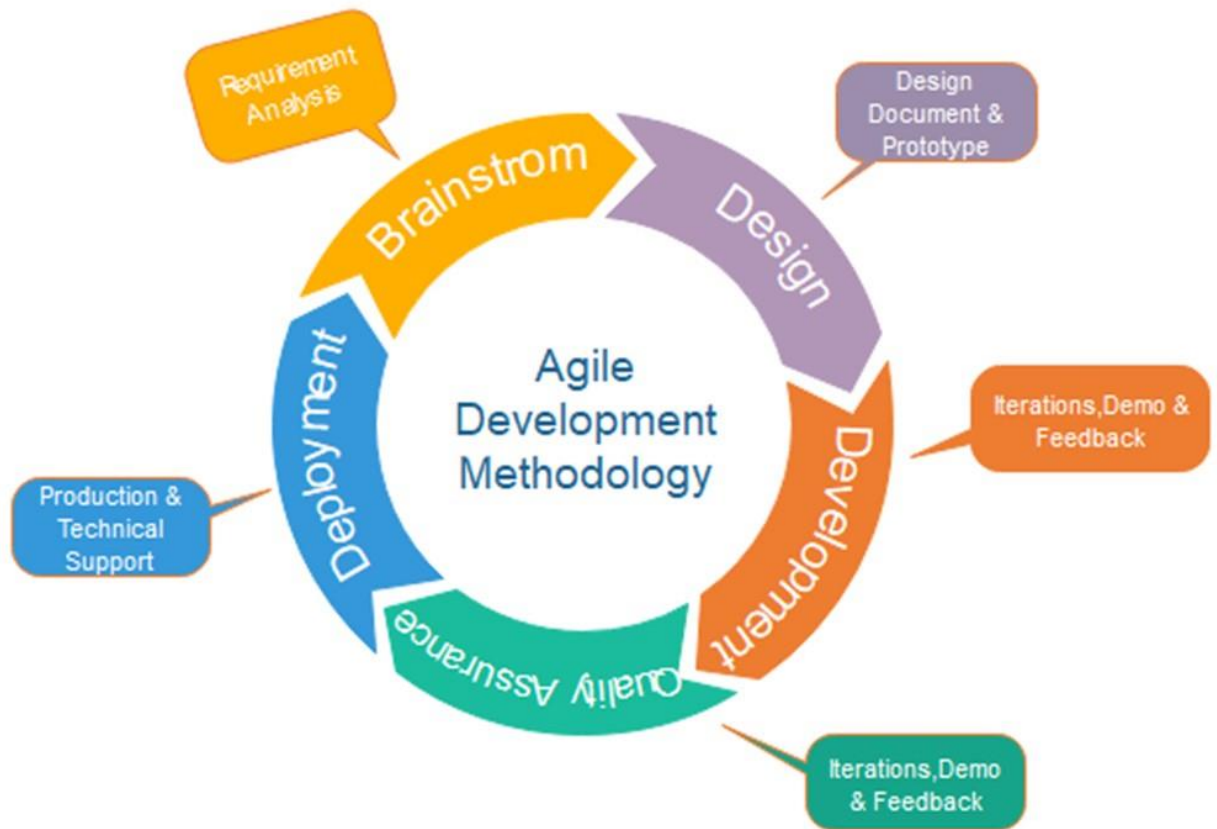


Fig. Agile Model

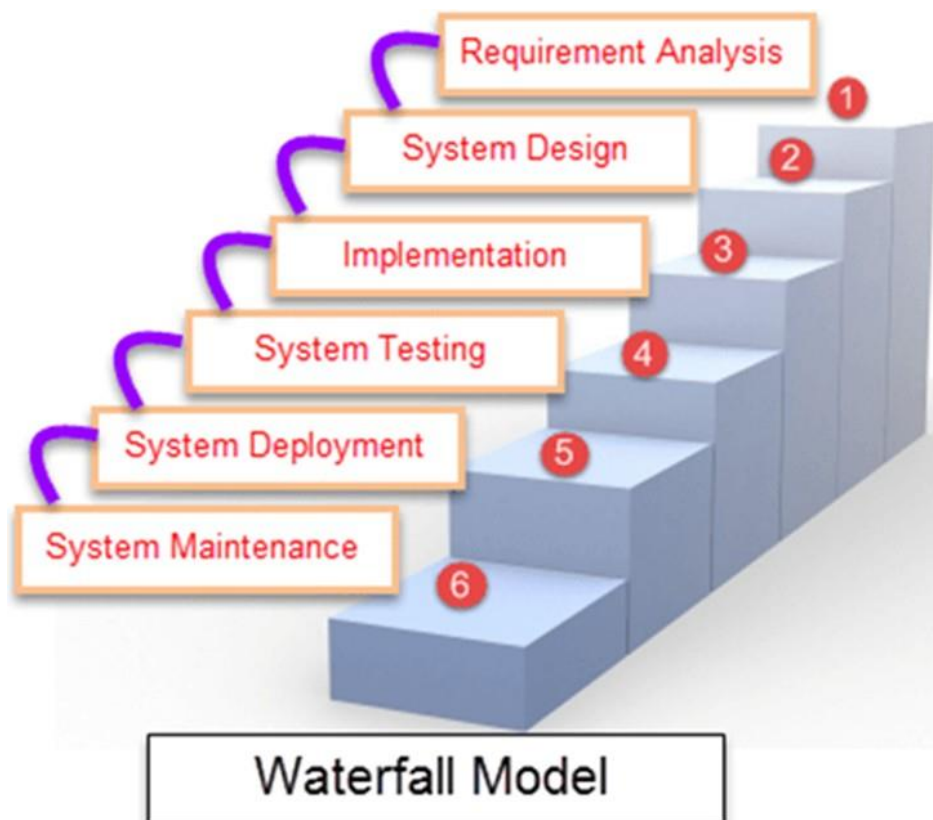
30-35% or even 50% of the IT projects fail according to research. Agile improves success rate by almost double by promoting better collaboration and communication. But Agile only provides a way to solve problems and how can I decide which is the right problem to solve? This is where Design thinking plays a vital role.

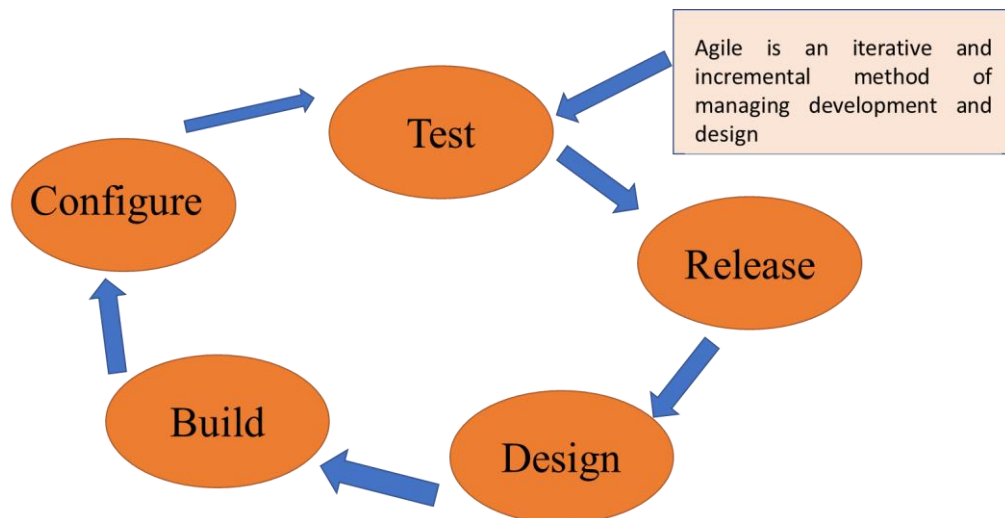
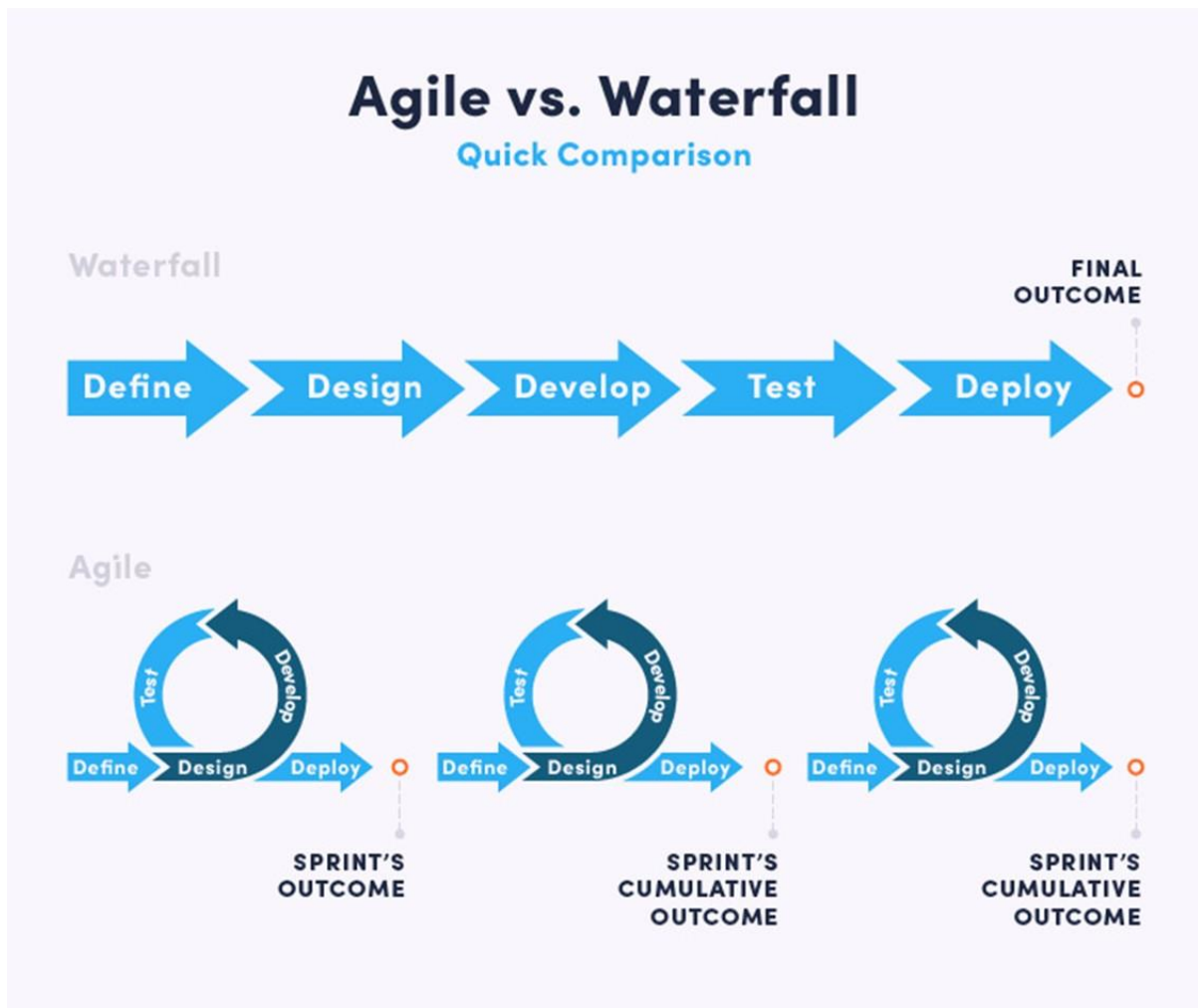
Combining both is not an easy task and requires a culture shift (new way of thinking and doing) and one has to get accustomed to this cultural shift. This combination values people over processes. Organizations have to allocate the right people for the projects and they should ensure cultural compatibility between teams and the way Design thinking and Agile methodology work.

Best practices for combining Design thinking and Agile methodology

1. Invest in user research
2. A clearly defined problem statement
3. Build a productive team culture
4. Optimal use of Design thinking
5. Design patterns and periodic testing

Waterfall Model





Advantages of Agile Development Method used in software development in IT Industry

- Solutions are Prototyped and Results are verified
- Best solutions are accepted
- Before approval itself the solutions are experienced by the client
- To improve user experience short iterations are possible
- Small cross-functional teams
- Incremental delivery is possible
- Fast feedback helps the designers and developers
- Continuous improvement is possible

AGILE IN VIRTUAL COLLABORATION ENVIRONMENT

In the book 'The Flat World' The author Thomas Friedman indicates the word 'FLAT' with an imaginary impression that technology and internet has changed the world so abundantly that in modern times people work remotely not only from a single geographic location but also across continents with different time zones, cultures and even languages.

Thomas Friedman explains flattening as (The world is flat, 2005) the combination of a personal computer with fiber optic micro cable, using the internet with the help of a working software. Working remote and in a global environment is not so uncommon anymore and has gained acceptance as a general working norm in the last couple of decades.

With insourcing, outsourcing and companies operating in a global environment do business globally. With the management headquartered at a single location however do operate with other regions with the use of technology. This means business run globally and hence projects within the organizations also path globally.

Agile methods are more popular in the software industry however have gain so much admiration that other industries also want to pursue its benefits in their businesses. Operating in a global environment makes these frameworks more challenging to operate efficiently.

Agile methodology can accept changes anytime compared to that of a waterfall method, and hence collaboration between collocated teams eases out the agile processes. Besides coordination; interaction, development, planning, review, retrospective sessions etc. also immensely reduce time and effort.

Working remote or distributed is considered being competitive and is considered a fit operating model. Some companies also provide it as a choice to its employees. Businesses want to leverage talent round the clock, use the best talent which may not be locally available, cheap labor by outsourcing to low cost countries, higher productivity and numerous strategic reasons.

With growing needs of working remote, it is intolerable to voice that agile methods will not work by distributed teams. Although few trials will arise, with the help of tools and techniques moving to a remote working environment is possible.

Benefits of Agile in Virtual Collaboration Environment

ALLOW TRANSPARENCY

As a company or a lead, establish a transparent environment. Provide a sense of trust in the team members with the decisions they make and organize sessions with them to align these decisions with company goals and vision. Do not keep responsibilities a secret. Be clear over job functions. Make sure the outcomes are public across the team. Gather feedback. Establish open communication channels

ESTABLISH A CULTURE OF CONTINUOUS IMPROVEMENT

Drive improvements within the team. Recognize potential improvement opportunities and obtain ideas from the team. Listen to suggestions and rationally take appropriate steps. Agile principle drives continuous improvement and thus team members are open for improvement opportunities. Generate calculated experiments to implement improvement initiatives.

COMMUNICATION

Communication is perhaps the most important artifact in any team. how you structure your communication will determine the output. Practice deep communication at all levels. A strong emphasis on communication is the key. Communication need not happen only via emails, phone calls or meetings, however in enterprises major communication happens with the help of ERP tools which help you be agile all the time.

CONSTRUCTION OF A RHYTHM

There are three things which are absolutely necessary in agile methodology: Transparency, inspection, and adaptation. Scrum also calls it the three pillars of Scrum (Scrum Guide, 2005). It is important we create a tempo which operates around these pillars of Scrum. Having said that this drives in creating a self-collaborating team which is a prime eminence of Agile.

GENERATE A CULTURE OF COURAGE & FLEXIBILITY

This also goes with the concept of failing fast. Failure is not bad; it is okay to fail. However, here the key argument is to fail fast. Establish an environment where the team is courageous to take

steps to try out something new. The idea is to reduce the delay. Detect the failure fast and further twist your initiatives.

ESTABLISH A SUSTAINABLE ENVIRONMENT AND WORK LIFE BALANCE

Often distributed teams end up stretching their normal working hours which lead to working more than regular hours or working at odd times. Establish rules and ensure no team member is defaulting on these principles, making the fellow counterpart abuse this principle. Plan appropriate and set expectations accordingly.

VISUALIZE EVERYTHING

A very important footstep. Planning of tasks is easier leverage tools that provide you a visual interpretation of tasks to be completed. Establish a solution that allows the team members to view workflows, depict who is working on what, align together and be able to pick the right task of right priority and at the right time. This solution will help you capture the right problem areas and effectively retrospect them to improve your processes.

SCENARIO BASED PROTOTYPING

Design thinking is at its best if tangible prototypes can be used to envision new products and services.

However, for complex software systems with multiple users usually such tangible prototypes are not feasible. To overcome this problem, scenario-based prototyping approaches can be proposed for designing complex software systems that are based on models, both structural and behavioral models.

The approach will support the stepwise and interactive enrichment of the prototype model, the traceability between the artifacts collected during earlier design phases and the scenarios. The models provide a more formal outcome of the design process for the downstream engineering activities, so that the gap between design and engineering can be narrowed.

Scenarios describe a sequence of events, illustrating the activities of one or more people in a real-world setting. The goals are to be realistic, detailed and concise. Since this is difficult to do quickly, it is best to cover only a limited period of time in the scenario. We distinguish between use scenarios, which are a form of analysis of the interviews and represent what happens today in real-world settings, and design scenarios, which are revised versions of use scenarios that illustrate how a new technology might be used.



Developing an app for IOS or Android and giving it to the customer for feedback Revisions are must

Instead of giving the original app developed, photos (Photoshop, illustrator or any other app) with real scenarios (push button enabled or directing to the test case) may be done. Now UI experts and User experience experts come to play and finally after many revisions the app is presented in its original form to the customer. (Even rolling out of Beta forms are in practice)

Types of Prototyping:

- Rapid prototyping/throwaway
- Evolutionary prototyping

Rapid prototyping/throwaway

Rapid prototyping (RP) quickly creates a physical part directly from its CAD model data using various manufacturing techniques. Rapid prototyping can be used at any stage of the product development cycle for any components or sub-components. Prototyping can be repeated numerous times along the new product design process using the test data to achieve the desired part.

Advantages of rapid prototyping

- Reduced design & development time
- Reduced overall product development cost
- Elimination or reduction of risk
- Allows functionality testing at a fraction of the cost
- Improved and increased user involvement during design stages of NPD

- Ability to evaluate human factors and ergonomics

Evolutionary prototyping

At what stage of the lifecycle is the Prototype; is it too close to the final product or does it require more iterations. Prototyping is a remarkable and quick way to bring ideas to life. Since the entire design thinking is an iterative process, the same gets applied to Prototype too. Once the Prototype is shown to other stakeholders, especially clients, their opinions and feedback can be taken into account to make iterations in the Prototype.

Creating a Prototype goes through a well-defined process that begins with making the Prototype, getting it reviewed, and finally refining it to match the expected product. A Prototype is generally classified into two broad categories: Low-fi, where a relatively simple version of Prototype is developed that can be tested immediately like a pen and paper drawing. On the contrary, a hi-fi version of Prototype is quite close to the final product and makes use of technology like a 3D plastic model with movable parts or an early version of a software system developed using a design program such as Sketch or Adobe Illustrator.

There are various methodologies and approaches to Prototype, like the Pen and Paper, Digital and HTML. Choosing the methodology entirely depends upon the objective, time, and cost implication. Before we wrap up on Prototype, there are a few guidelines to keep in mind before creating a Prototype:

- Not clear as to what to make, then start making: This is the best bet if you are unsure of what needs to be created, it is better to start creating and gaining meaningful insights from it.
- Time saved is money earned: Time is crucial and doesn't spend too long creating a Prototype as it might have a considerable cost implication.
- Practice speed: Agree or not; it is the key. The more agile and speedy you are, the closer you are to a successful Prototype.
- Keep in mind what you are testing: If Prototype has become cumbersome, you are most likely to deviate from your objective. Do not lose sight of your purpose and put all your efforts in the same direction.
- Keep the user in mind: Last but not least, do not lose your focus on the end-user; he is the one for whom the entire exercise is done!